

Question 1: Scan the Local Wi-Fi Network

Command Used:

```
nmap -sn 192.168.184.0/24
```

The Nmap scan identified three active devices on the local network. Two devices (192.168.184.2 and 192.168.184.254) were detected as VMware devices based on their MAC address vendor, while 192.168.184.130 is my computer. All three devices were online and responded during the network scan.

Q 2

Answer:

The `arp -a` command was used to display the Address Resolution Protocol (ARP) table, which contains the IP addresses and corresponding MAC addresses of devices recently communicated with on the local network. The output identified the default gateway with IP address **192.168.184.2** and MAC address **00:50:56:F6:E8:1F**. At the time of the scan, this was the only active device present in the ARP cache.

Table:

Device	IP Address	MAC Address
Default Gateway	192.168.184.2	00:50:56:F6:E8:1F

Q 3

Port Number	Service	Status
22	SSH	Closed
80	HTTP	Closed
443	HTTPS	Closed

The Nmap scan showed that ports **22 (SSH)**, **80 (HTTP)**, and **443 (HTTPS)** were **closed**. No common network services were running on the local computer at the time of the scan.

Q 4

Device	IP Address	MAC Address
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The Nmap scan of **scanme.nmap.org** identified four open ports. Port **22** is running the **SSH (OpenSSH 6.6.1p1)** service, which is commonly used for secure remote login. Port **80** is running **Apache HTTP Server 2.4.7**, indicating that the server hosts a website. Port **9929** is running the **Nping Echo** service, which is used by Nmap for network testing. Port **31337** is reported as **tcpwrapped**, which means the service is protected or restricts information disclosure. These services are intentionally available because **scanme.nmap.org** is the official Nmap test server.

Q 5

The Nmap scan was performed using the command `nmap --open -sV scanme.nmap.org`. The `--open` option displayed only the open ports, while the `-sV` option detected the services and their versions. The scan identified four open ports: 22 running OpenSSH 6.6.1p1, 80 running Apache HTTP Server 2.4.7, 9929 running the Nping Echo service, and 31337 reported as tcpwrapped. Using `--open` made the output cleaner by hiding all closed ports and showing only active services with their versions.